

regulatory sequences and other functions.

The next step is to determine the function of a gene by studying gene expression, which will establish when, where and how much of a protein is produced. These processes are assisted by comparing one DNA sequence with the sequence of other similar organisms with the help of sophisticated computers.

Bioinformatics services

As databases grow bigger, higher computing power is required and hence cluster-type computers serve the purpose better. Since the entire database is not likely to be stored in one place but distributed across the globe, and researchers worldwide will need to access it, grid-style computing is the solution. BioPerl (for solving high-level tasks) and EMBOSS (for compiled programs) are some well-known grid-style computing projects.

BioPerl. BioPerl project is a co-ordinated initiative to collect computational methods routinely used in bioinformatics and life-science research into a set of standard CPAN-style, well-documented and widely-available Perl modules.

Perl is useful for a variety of problems. Before this project, Perl provided unparalleled support for many jobs common in bioinformatics and life-science research; however, there were no standard Perl modules for biology.

BioPerl targets to fill this gap by convincing scientists and experts to share more of their own codes on public domains for others to see, learn from, use and improve. BioPerl owes its early organisational support to its association with VSN-BCD bio-computing courses.

BioPerl was later adopted by people developing codes at large genome centres such as Stanford and Washington universities. It was in 2002 that BioPerl 1.0 (a significant release since incorporation), a stable and easy-to-use toolkit, was launched.



Fig. 3: Human Genome Project is an international research effort to determine the sequence of the human genome and identify the genes it contains (Image courtesy: cancer.org/research/what_we_know/omics/genomics_04.html)



Fig. 4: BioPerl is a collection of Perl modules that facilitate the development of Perl scripts for bioinformatics applications. It has played an integral role in Human Genome Project (Image courtesy: www.lbiexperts.com)

One of the first-generation server hardware for bioperl.org (and other open-bio.org-hosted projects) was provided by Compaq Computer Corp.

Today, BioPerl is a flourishing open source project, with nearly all of the founding developers still active. BioPerl modules and source codes are freely available under Perl, especially critical for researchers and lab technicians who lack the resources to hire dedicated software engineer teams. The project is open to all and invites suggestions and participation from the larger bioinformatics community.

EMBOSS

EMBOSS, or The European Molecular Biology Open Software Suite, is a free open source software analysis package specially developed for the needs of the molecular biology (for example, EMBnet) user community. The software automatically deals with data in a variety of formats and

also facilitates transparent retrieval of sequence data from the Web.

Since extensive libraries are provided with the package, it enables other scientists to develop and release software in true open source spirit. EMBOSS also integrates a gamut of currently-available packages and tools for sequence analysis into a continuous whole. Some application areas of EMBOSS are sequence alignment, rapid database searching with sequence patterns, protein motif identification, nucleotide sequence pattern analysis, codon usage analysis for small genomes, rapid identification of sequence patterns in large-scale sequence sets and presentation tools for publication, apart from other application areas.

EMBOSS has several important benefits. First, it has a suitably-built toolkit for developing strong bioinformatics application or workflow. There is a comprehensive set of sequence-analysis programmes, and all sequence and several alignment and structural formats are managed.

Then, there is an extensive programming library for common sequence-analysis tasks and additional programming libraries for tasks such as string handling, pattern matching, list processing and database indexing.

EMBOSS works on almost every UNIX, Windows and Mac operating system. Since each application has the same style of interface, mastering one means mastering all. The consistent user interface facilitates GUI designers and developers. EMBOSS is free of arbitrary size limits, that is, there are no limits on the amount of data that can be processed. And for programmers, memory management for objects such as sequences and arrays is simplified.

AI and bioinformatics

AI and heuristic methods are very critical for the evolution of bioinformatics. Presently, molecular biologists are involved in some no-